Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/13/2025
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Spaceflight Effects on Plant-Microbe Interactions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	04/01/2021	**End Date:**	09/30/2025
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2025 per H. Levine/KSC (Ed., 2/24/25). NOTE: End date changed to 03/31/2025 per NSSC information (Ed., 4/8/24).		
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Bakshi, Arkadipta Ph.D. (University of Wisconsin, Madison) Swanson, Sarah Ph.D. (University of Wisconsin, Madison) Barker, Richard Ph.D. (University of Wisconsin System) Hanson, David Ph.D. (University of New Mexico)		
Grant/Contract No.:	80NSSC21K0577		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work seeks to address: (1) how spaceflight modulates the interactions between plants and microbes and (2) how well microgravity analogs capture the events elicited by the spaceflight environment. Tomato plants will be grown on orbit in the NASA Vegetable Production System (Veggie) hardware on board the International Space Station (ISS) with and without the beneficial rhizosphere microbe Trichoderma hazianum. A third sample will be of this microbe growing under identical conditions on the ISS but without the plants. These samples will be compared to parallel ground controls at 1 x gravity as well as to samples growing on 1-axis and 3D clinostats. Assays will integrate RNAseq-based transcripomics and ionomics (nutrient uptake and distribution) alongside biochemical measures of photosynthesis and stress. These comparisons will provide measures of both how spaceflight affects the plant, the microbe, and the relationship between these organisms and additionally, how well microgravity analogs can reproduce these kinds of events on the ground. In addition, the omics-level data gathered from this study will be compared to the wealth of spaceflight-related omics data available through the GeneLab data repository. Using an approach of orthologous matrix mapping will allow identification of similar genes between diverse species and so allow for comparisons of, for example, the degree of similarity between patterns of gene expression to be compared between different species. Overall this research will help define how spaceflight may modulate plant, microbial, and plant-microbe responses and help understand whether defined beneficial microbes may provide a countermeasure to the deleterious effects of spaceflight on plants. The work will capitalize on the complementary expertise of two groups: the Gilroy lab team (plant spaceflight, transcriptomics) and the Hanson lab (biochemistry, photosynthesis).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed research seeks to address how spaceflight modulates the interactions between plants and microbes using tomatoes and the beneficial soil microbe Trichoderma harzianum. The microbiome around the plant root is recognized as a crucial element in the productivity and hardiness of plants but the complex interactions and chemical signals that occur between plant and microbe have only recently begun to be dissected. Yet, for example, T. harzianum is used as a commercial biostimulant, being added to the soil to pormote plant growth and vigor. The research in this project will further define the molecular components of the interaction between plant root and this fungus and how spaceflight alters these events. Thus, the work will not only provide insight into how plant-microbe interactions are affected by spaceflight but address whether T. harzianum might be used as a biostimulant to counteract some of the stresses of spaceflight on plant growth. This research will also help provide molecular insight into how these interactions occur on Earth. Such a fuller understanding of plant-microbe interactions, especially of T. harzianum-plant root communications and response will be important steps towards optimizing these beneficial interactions and so increasing plant productivity in both space and on Earth.
Task Progress:	FLIGHT EXPERIMENT: The experiment (Advanced Plant Experiment 10) was launched on the NG-20 Cygnus spacecraft. All in orbit operations were successfully completed in the Veggie hardware on the International Space Station and all the biology samples were returned to Earth on the CREW 7 vehicle. [Ed. Note: The NASA Vegetable Production System (Veggie) is a plant growth unit on the International Space Station (ISS). It is capable of producing salad-type crops to provide the crew with a safe, palatable, and nutritious source of food, while also providing a tool to support recreation and relaxation.] The biological samples consisted of sterile Petri plates containing nutrient gel medium with either tomato seeds alone, tomato seeds inoculated with spores of T. harzianum, or Petri plates inoculated with fungal spores alone. A 2 mm pore size polyester mesh that was bounded by a 3 mm wide, 0.5 mm thick, 3d printed PETG perimeter rim was incorporated onto the top of the nutrient gel before planting of the seeds or inoculating with fungal spores to facilitate crew harvesting of plants and fungal colonies at the end of the experiment. Plants were grown under continuous light for 14 days in the Veggie hardware, with crew photography every other day. The experiment also included gas sampling of the air around the Veggies to allow stable carbon isotope analysis for calculation of photosynthetic and physiological parameters such as plant water use efficiency. On day 14, plates were opened in the Life Sciences Glovebox on the ISS, to ensure fungal containment, and biological materials were harvested, frozen between aluminum blocks conditioned to -160˚C, and stored in the Minus Eighty (Degrees Celsius) Laboratory Freezer for ISS (MELFI) for sample return. On orbit data included imaging of the plants and fungi as they grew and these images reveal that plants showed shorter roots and enhanced lateral root production in spaceflight when compared to ground controls. Treatment with Trichoderma led to reduced primary root growth and increased root system branching with high growth rates of the Trichoderma. Cabin air was also sampled for comparison to an analysis of stable carbon isotope ratios of the biological materials and analysis indicates that the 13C:12C isotopic ratio is suppressed in spaceflight although carbon to nitrogen content is unchanged. Biochemical analysis of the samples has included measurement of chlorophyll and carotenoids as indicators of photosynthetic pigment content, quantification of the stress pigment anthocyanin and measurement of the levels of malondialdehyde, a biochemical reporter of oxidative stress. These analyses have revealed that although spaceflight did not alter photosynthetic pigment accumulation, it did lead to a shoot-level signature of stress (significant anthocyanin and malondialdehyde (MDA) accumulation). Further analysis for these samples includes transcriptomic analysis via RNAseq and glycomic profiling of the cell wall materials. Glycomics involves extracting cell walls and then using a panel of monoclonal antibodies against known wall epitopes to quantify wall polymer abundance using an Enzyme-Linked Immunosorbent Assay (ELISA)-based assay. Glycomic profiling has now been optimized for use with these spaceflight-related samples across multiple plant species. CURRENT WORK: Current work is focused on completing the full glycomic profile from the Advanced Plant Experiment 10 (APEX 10) spaceflight samples and a set of parallel clinostat grown tomato samples, along with ionomic profiling. Ionomics quantifies trace element composition of 15 minerals within the plant using Inductively Couple Plasma Mass Spectroscopy. PRESENTATIONS AND OUTREACH: These spaceflight-related projects continue to form a key part of the laboratory’s outreach efforts. They have been presented at venues ranging from the meeting of the International Space Life Science working Group in Liverpool, UK, to the American Society for Gravitational and Space Research, to the Plant Cell Dynamics meeting, to the Midwest sectional meeting of the American Society of Plant Biologists. This research has also formed the core of outreach to the

	general public at events including the University of Wisconsin's Science Expeditions and the 'Space Camp' offered to 5th-8th graders at the Deke Slayton Space Museum in Sparta, Wisconsin. Interviews about the flight portion of this grant have been presented to media such as Wisconsin Public Radio, WORT radio, The Midwest Farm Report, Spectrum News and local new outlets and magazines such as the Wisconsin State Journal.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Bakshi A, Gilroy S. "Calcium signaling in hypoxic response." Plant Physiol. 2024 Dec 24;197(1):kiae654. https://doi:10.1093/plphys/kiae654 ; PubMed PMID: 39707915 , Dec-2024